



Project Report

Project Title: **Space Aircraft**

Name : Rafiul Omar Rafi, ID: 2233192042

Name : Md. Adnan Wasti, ID: 2233693642

Name : Tahmid Anjum Rifat, ID: 2233056642

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Introduction :

Space Aircraft is a simple “shoot ‘em up” type game. In this game, you will pilot your own spaceship and battle against waves of enemies.

The rules of this game are simple: Use arrow keys or w/a/s/d to move your ship. Use the “spacebar” to shoot. Avoid enemy ships, keep moving to avoid collisions.

This game has no end. You can progress indefinitely until you die(collide with any enemy ship). You get one life. Though this game has no end, it has a level system to increase difficulty gradually. Each level has five more enemies than the previous one. You have to destroy all the enemy ships of a certain level to progress to the next level.

The ultimate goal of this game is to beat your friend’s high score. Yes, this game has a scoring system that records your highest score. Three different player profiles can be set and their names will be saved along with their scores.

This game is a great example of how to use Java to create a simple but fun game. The game is well-written and easy to understand, and it makes use of a variety of Java features, such as object-oriented programming, ArrayList, and graphics 2D. The game is also well-designed and fun to play.

Libraries Used:

In this Game, we have used

- Java Swing (panel, paint, button, combobox, radiobutton)
- Java Graphics 2d (For painting the image, and shape in the game panel)
- Timer (For the game running thread)
- Action Event (For button pressing)
- KeyAdapter (For key listening)

External resources: Media And Resources from online (free to use).

During the making of the game, we implemented all the OOP features.

Project Features Table:

Java Class	Feature & Description	Implemented by
AirCraft	Java Frame, which hold the game panel	Md. Adnan
Entity	Set the coordinate of Image Ship	Tahmid Anjum
Enemy	The enemy of the game, The class is extends the entity class	Tahmid Anjum
GameFrame	Game panel, which hold all the game items runnable, and used thread	Md. Adnan
KeyAdapt	Key class a custom class extends KeyAdapter, which will control the player of the game.	Md. Adnan
Missile	The Missile of the game, The class extends the entity class. The player can shoot the missile to destroy the enemy ship.	Tahmid Anjum
Player	The player of the game, The class extends the entity class. The player ship has the abilities to perform moves and shoot the enemies.	Rafiul Omar and Tahmid Anjum
Score	The score class is responsible for displaying the score in the game panel.	Md. Adnan

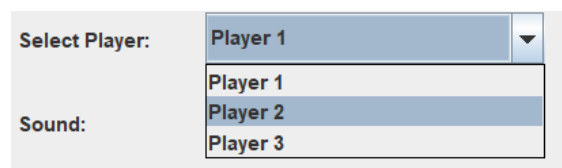
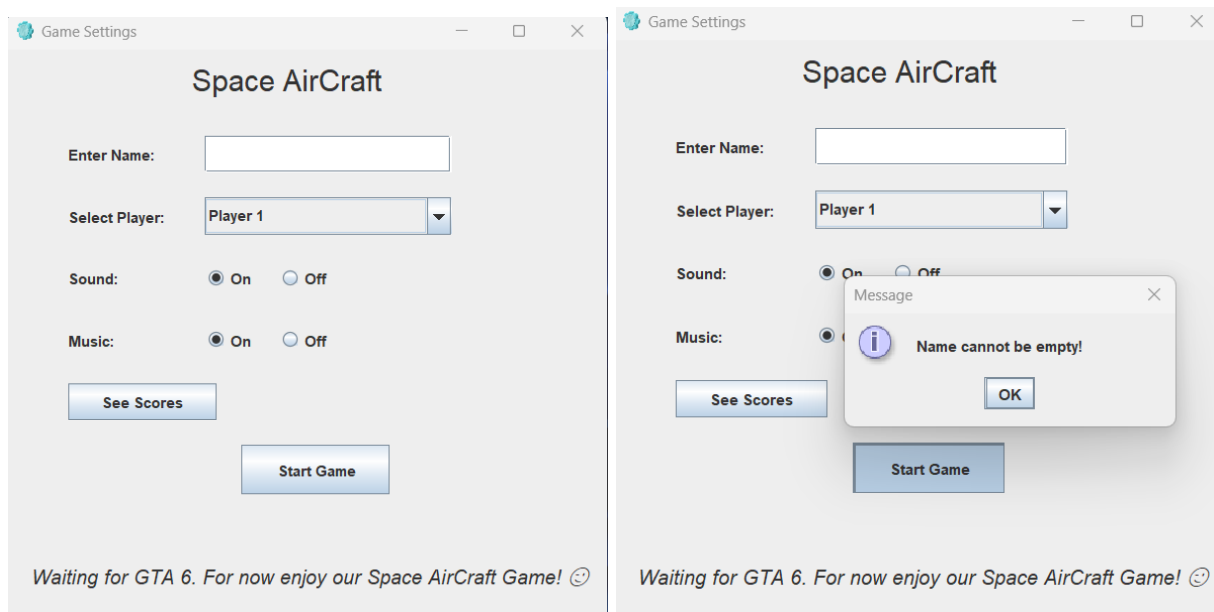
Sound	The sound class has all the file operation and sound for the game. Missile sound, game music, and enemy destroy sound.	Rafiul Omar
Data	Data class is for how the name and score are stored in the file.	Rafiul Omar
DataWindow	All the score and name are show in the new frame from the data.txt file	Rafiul Omar
NameBlankException	Custom exception to check the name field error.	Rafiul Omar
Settings	The setting class is responsible for changing the game setting. The class implements the SettingsUtility.	Rafiul Omar
SettingsUtility	This is the interface.	Rafiul Omar
Main	The Main class where the game should start.	Tahmid Anjum

Any of the image and sound files are not ours, we took them from various websites which are free to use.
The file data.txt is used for storing all the names and scores.

Description:

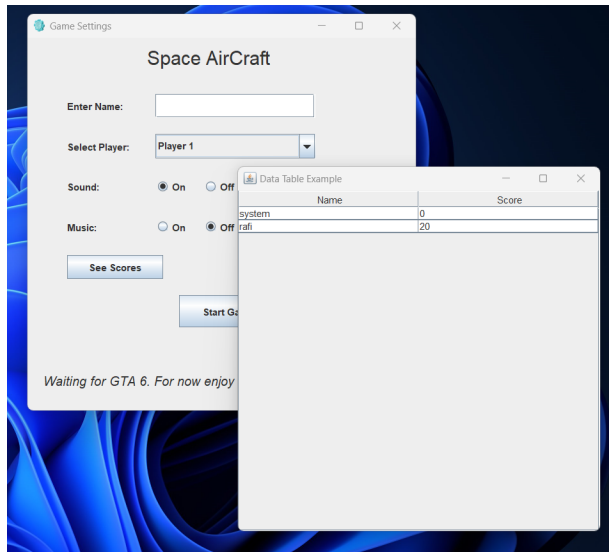
The game's settings panel:

After starting the game, a user will be presented with the following screen.

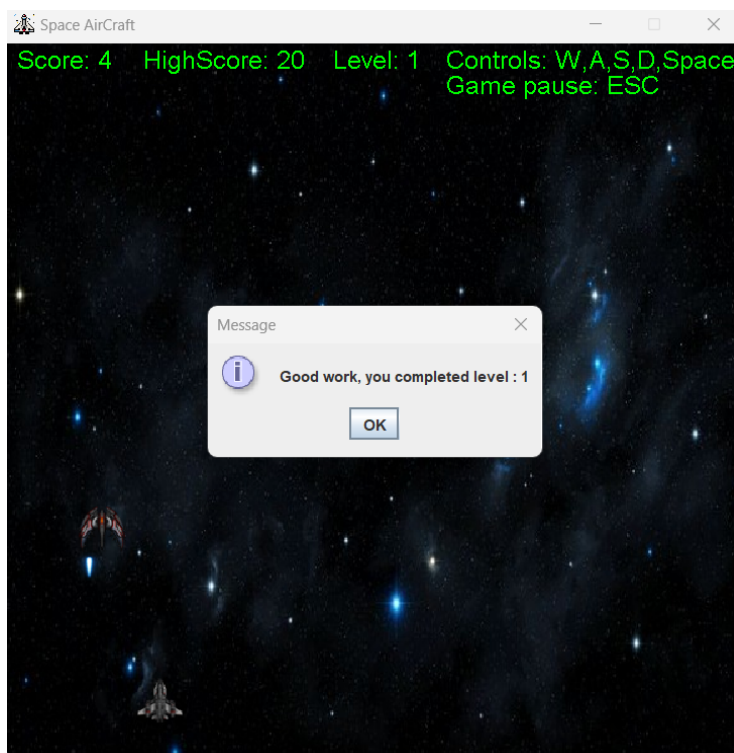


From the above images, it can be seen that a user is asked for his name first. This is to save his score later. As without a name any score can not be specified to anyone, thus a user is not let to start the game without setting his name first.

Next, the user has to select a player profile. Three profiles can be chosen. There are also options for switching sound and music off and on.



There is a score button on the panel too. By pressing the button, a player can see his highest score and also the highest scores of other players.



This image shows the leveling system. By clearing all the enemy ships in a level, the player gets to the next level. Each level has five more enemies from the previous level. So the more a person plays, the harder it gets.



This image shows the player ship shooting at multiple enemy ships. This is done by pressing the spacebar. “Align your ship with an enemy ship, press the spacebar, bullet hits the enemy ship, enemy ship destroyed” simple mechanism.



This image shows how the player dies or the game ends. This simply happens when the player ship collides with the enemy ship. After that, the score of that player will be saved and they have to start over.

Some problems that we faced during making the game:

- The file handling and storing data in the data.txt file, and getting all the data and calculating the high score from the game start.
- In the settings class some methods need to be static, before the game encounters bugs.
- In the Setting class we had to face several problem to customize the music and sound file with conditions and input fileld.

Challenges we faced:

- Learning and implementing from scratch and done within a short time.
- Group task divided and learned new game concepts and game resources from online.
- Making the player and enemy bound on the screen.
- Check collision for the player and the enemy.

For the future plan, update, and implementations:

- We will create enemy bosses for certain levels.
- We will implement custom enemies for each level
- We will add some booms and sceneries for each level.

Conclusion:

The game 'Space Aircraft' is a perfect example of a project using all the basic libraries of Java. This is a fun game, which has no specific end until the user Airplane crashes. The main target of the game is breaking your friends' records, sometimes your own previous records. These projects also helped us to learn how to do teamwork perfectly. Getting help from the team partners when one faces any difficulty is one of the memorable moments during the projects.